

Contestant Number: _____

Time: _____

Rank: _____



**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential

DATABASE APPLICATIONS (240)

STATE 2022

Production:

Job 1: Sportswear Table _____ (100 points)

Job 2: School Table _____ (100 points)

Job 3: Order Table _____ (100 points)

Job 4: Invoice by School Report _____ (100 points)

Job 5: School Mailing Labels _____ (50 points)

TOTAL POINTS _____ (450 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

DATABASE APPLICATIONS SCORING SHEET

<i>Unless indicated otherwise, student receives all points or none.</i>	Points Possible	Points Earned
Job 1 - Sportswear Table (100 points maximum)		
Item ID, Description, On Hand, Reorder Level, and Markup are designed as instructed (earn 10 points per field for accurate field names, appropriate data types, and all appropriate properties set-including design notes)	60	
Order? is a calculated field and displays the text Yes/No if Reorder Level is greater than On Hand.	15	
Proofreading/data entry (.5 points off for each error)	15	
Sorted by Description (ascending)	5	
Printed using best fit in portrait orientation (all data shows, 1 page)	5	
Total Points - Job 1	100	
Job 2 - School Table (100 points maximum)		
Designed as instructed (earn 5 points per field for accurate field names, appropriate data types, and all appropriate properties set-including design notes)	45	
Proofreading/data entry (.5 points off for each error)	20	
Appended the table for a total of 13 records	25	
Sorted by the primary key (ascending)	5	
Printed in landscape orientation (all data shows, 1 page)	5	
Total Points - Job 2	100	
Job 3 - Order Table (100 points maximum)		
Imported 15 records from the "Orders to Import" Excel workbook	10	
Design is updated for Order Date, Cost, and Quantity (earn 10 points per field for accurate field names, appropriate data types, and all appropriate properties set-including design notes)	30	
Order ID added (primary key, AutoNumber)	10	
Item ID looks up values from the Sportswear table with both the Item ID and Description shown in the list.	10	
School ID looks up values from the School table with both the Vendor ID and Vendor Name shown in the list.	10	
Shipping field added (Quantity * Cost * 10%)	10	
Invoice Total field added (Cost * Quantity + Shipping)	10	
Sorted by Order Date (oldest to newest)	5	
Printed using best fit in landscape view (all data shows, 1 page)	5	
Total Points - Job 3	100	
Job 4 – Invoice by School (100 points maximum)		
Correct fields are visible: School Name, Address, City, State, Zip, order date, cost, quantity, shipping and Invoice Total (5 points each)	55	
Subtotal of Invoice Total per School with a grand total	20	
All Columns on one page	5	

Grouping correct	10	
Amounts match the sample provided	5	
Printed in Landscape orientation, (all data shows, 1 page)	5	
Total Points - Job 4	100	
Job 5 – School Mailing Labels		
Used Avery C2163 mailing label	10	
Set Font to Times New Roman	5	
Set Font size to 12	5	
Label Set up correct	10	
Sorted by ZIP	10	
Printed	10	
Total Points – Job 5	50	
Total Points Earned	450	

Create a database named **State-XXXXX**, and save. (XXXXX is your Member ID)

Job 1: Sportswear Table

1. Create the table named **Sportswear XXXXX** where XXXXX is your Member ID. Use the same field names and assign best data types and properties considering the field values shown below. Include the notes listed below the table in your design.

Sportswear XXXXX					
Item ID	Description	On Hand	Reorder Level	Order?	Mark Up
HR-001	Hoodie-Red	57	50	Calculated Field	25%
HB-002	Hoodie-Blue	21	50	Calculated Field	25%
ZR-001	Half-Zip-Red	32	25	Calculated Field	15%
ZB-002	Half-Zip-Blue	40	25	Calculated Field	15%
TR-001	Cotton T-Shirt-Red	45	50	Calculated Field	20%
TB-002	Cotton T-Shirt-Blue	56	50	Calculated Field	20%
LR-001	Long Sleeved T-Shirt Red	23	10	Calculated Field	30%
LB-002	Long Sleeved T-Shirt Blue	8	10	Calculated Field	30%

Design notes:

Field Name	Property Notes
On Hand	- Integer Field Size - Fixed Format 0 Decimal Places
Reorder Level	Integer Field Size
Order?	- Use a calculation for this field. Do not type in the words. - If the Reorder Level is greater than On Hand, display “Yes.” If not, display “No.” - Adjust properties as needed to show words in the field, not numbers.
Markup	- Double Field Size - Percent Format 0 Decimal Places

2. Enter all the records.
3. Adjust the column widths so all entries are visible.
4. Sort the table by Description in ascending order.
5. Print the table in portrait orientation. All data should be visible on one page.
6. Use database Documenter to verify fields/types/properties. Print it.

Job 2: School Table

1. Create the table named **School XXXXX** where XXXXX is your Member ID. Use the same field names and assign best data types and properties considering the field values shown below. Include the notes listed below the table in your design.

School XXXXX								
School ID	School Name	Address	City	State	ZIP	Phone	Discount Terms	Net Terms
TC47995	Tri-County High School	341 HWY 24	Remington	IN	47977	(219) 555-5800	3%	45
BR47362	Blue River Valley	4741 Viking Trail	New Castle	IN	47362	(293) 555-1111	2%	30
EC47710	Evansville Central High School	5400 N First Ave	Evansville	IN	47710	(812) 555-5000	10%	30

Design Notes:

Field Name	Properties
Phone	Set an appropriate input mask and store the symbols
Discount Terms	- Single Field Size - Percent Format - Zero Decimals
Net Terms	Single Field Size

2. Enter all the records.
3. Best fit column widths.
4. Append table with the excel worksheet named School.
5. Sort the table by the primary key in ascending order.
6. Print the table in landscape orientation. All data should be visible on one page.

Job 3: Order Table

1. Import the table named **Order XXXXX** where XXXXX is your Member ID from the Excel workbook "Order" on the storage device you were provided. Your table should have 18 records.
2. Review the records. Update the table design to use best data types and properties considering the field values. Include the following notes in your design:

Field Name	Properties
Order ID	Add this auto numbered field for the primary key.
Item ID	Look Up these values from the Sportswear table. Display both the Item ID and Description in the drop down list.
School ID	Look Up these values from the School table. Display both the School ID and School Name in the drop down list.
Quantity	Integer Field Size
Cost	Currency

3. Add the following calculated fields:
 1. Shipping (Quantity * Cost * 10%)
 1. Format Currency
 2. Invoice Total (Invoice Total +Shipping)
 1. Format Currency
4. Best fit column widths.
5. Sort the table by Order Date (oldest to newest).
6. Print the table in landscape orientation. All data should be visible on one page.

JOB 4: INVOICE BY SCHOOL REPORT

1. Create a report using the school table and the orders table.
2. Name Report Invoice by School XXXXX where XXXXX is your Member ID.
3. Group by School.
4. Landscaped with all columns on one page.
5. Include fields in picture below.
6. Sum Invoice Totals with a grand total at the end.
7. Arrange the fields and resize columns as needed. Your report should look similar to the following example:

Invoice by School XXXX									
School Name	Address	City	State	ZIP	Order Date	Cost	Quantity	Shipping	Invoice Total
Ardmore High School	30285 Ardmore Ave	Ardmore	AL	35739	2/15/2021	\$35.75	500	\$1,787.50	\$19,662.50
					8/19/2021	\$10.00	50	\$50.00	\$550.00
					1/19/2021	\$35.75	50	\$178.75	\$1,966.25
Sum									\$22,178.00
Avail School	425 C Street	Anchorage	AK	99501	4/1/2021	\$10.00	40	\$40.00	\$440.00
Sum									\$440.00
Beta	4680 Lake Underhill R	Orlando	FL	32807	3/5/2021	\$10.00	50	\$50.00	\$550.00
					2/28/2021	\$35.75	25	\$89.38	\$983.13
Sum									\$1,533.00
Booker High School	3201 N Orange Ave	Sarasota	FL	34234	4/1/2021	\$38.00	10	\$38.00	\$418.00
					2/13/2021	\$15.00	15	\$22.50	\$247.50
Sum									\$665.00
Cleveland Early Colleg	2075 Stokes Blvd	Cleveland	OH	44106	4/15/2020	\$35.75	100	\$357.50	\$3,932.50
					10/2/2021	\$38.00	100	\$380.00	\$4,180.00
Sum									\$8,112.00
Dover High School	One Dover High Drive	Dover	DE	19904	1/2/2021	\$35.75	50	\$178.75	\$1,966.25
					4/30/2020	\$15.00	50	\$75.00	\$825.00
Sum									\$2,791.00
Danville High School	9235 Danville Rd	Danville	AL	35619	7/30/2020	\$10.00	50	\$50.00	\$550.00

JOB 5: CREATE SCHOOL MAILING LABELS

1. Use the Label Wizard to create mailing labels for the School table, and include the following options:
 - a. Use Avery C2163 as the label size.
 - b. Use Times New Roman with a font size of 12. Font weight normal.
 - c. On the first line of the label, include the School Name Field.
 - d. On the second line of the label, include the Address field.
 - e. On the third line of the label, include the City field, a comma, a space, the State field, a space, and the ZIP field.
 - f. Sort by Zip.
 - g. Name Labels School XXXXX where XXXXX is your Member ID.
2. Your label should look similar to the partial one shown below.
3. Print

Dover High School One Dover High Drive Dover, DE 19904	Beta 4680 Lake Underhill Rd Orlando, FL 32807
Booker High School 3201 N Orange Ave Sarasota, FL 34234	Danville High School 9235 Danville Rd Danville, AL 35619
Ardmore High School 30285 Ardmore Ave Ardmore, AL 35739	Cleveland Early College High School 2075 Stokes Blvd Cleveland, OH 44106